

1. File Descriptors

```
hecker@Windows: ~/Project_Os/Week_3
hecker@Windows:~$ cd Project_Os
hecker@Windows:~/Project_Os$ mkdir Week_3
hecker@Windows:~/Project_Os$ cd Week_3
hecker@Windows:~/Project_Os/Week_3$ echo -e '#!/bin/bash\n echo "normal output"\nls wrongfile' > test.sh
hecker@Windows:~/Project_Os/Week_3$ chmod +x test.sh
hecker@Windows:~/Project_Os/Week_3$ ./test.sh
normal output
ls: cannot access 'wrongfile': No such file or directory
hecker@Windows:~/Project_Os/Week_3$ ./test.sh out.txt
normal output
ls: cannot access 'wrongfile': No such file or directory
hecker@Windows:~/Project_Os/Week_3$ ./test.sh 2> err.txt
normal output
hecker@Windows:~/Project_Os/Week_3$ ./test.sh > all.txt 2>&1
hecker@Windows:~/Project_Os/Week_3$ ./test.sh > out.txt 2> err.txt
hecker@Windows:~/Project_Os/Week_3$ cat out.txt
normal output
hecker@Windows:~/Project_Os/Week_3$ cat err.txt
ls: cannot access 'wrongfile': No such file or directory
hecker@Windows:~/Project_Os/Week_3$ cat all.txt
normal output
ls: cannot access 'wrongfile': No such file or directory
hecker@Windows:~/Project_Os/Week_3$
```

2. Pipeline

```
hecker@Windows: ~/Project_Os/Week_3
hecker@Windows:~/Project_Os/Week_3$ echo -e "error: disk\ninfo: ok\nerror: disk\nerror: cpu\ninfo: ok\nerror: cpu" > log.txt
hecker@Windows:~/Project_Os/Week_3$ grep error log.txt
error: disk
error: disk
error: cpu
error: cpu
hecker@Windows:~/Project_Os/Week_3$ grep error log.txt | sort
error: cpu
error: cpu
error: disk
error: disk
hecker@Windows:~/Project_Os/Week_3$ grep error log.txt | sort | uniq -c
  2 error: cpu
  2 error: disk
hecker@Windows:~/Project_Os/Week_3$ grep error log.txt | sort | uniq -c | sort -nr
  2 error: disk
  2 error: cpu
hecker@Windows:~/Project_Os/Week_3$ grep error log.txt | sort | uniq -c | sort -nr | awk '{print $2, $3, "=", $1}'
error: disk = 2
error: cpu = 2
hecker@Windows:~/Project_Os/Week_3$
```

3. Heredoc

```
hecker@Windows: ~/Project_Os/Week_3
hecker@Windows:~/Project_Os/Week_3$ cat > script.sh << 'EOF'
#!/bin/bash
while read line
do
    echo "Processed: $line"
done
EOF
hecker@Windows:~/Project_Os/Week_3$ chmod +x script.sh
hecker@Windows:~/Project_Os/Week_3$ ./script.sh << EOF
apple
banana
cat
EOF
Processed: apple
Processed: banana
Processed: cat
hecker@Windows:~/Project_Os/Week_3$
```